

Ex No. 12 Implementation of Repository Pattern with WorkManager in Android**Aim**

To implement the Repository Pattern for data handling and use WorkManager to perform background tasks in an Android application.

Objectives

- Understand Repository Pattern in Android architecture
- Perform background tasks using WorkManager
- Separate UI logic from data handling logic
- Execute reliable background operations

Repository Pattern

- Acts as a **single source of data**
- Separates **UI layer** from **data operations**
- Improves code maintainability and testability

WorkManager

- Jetpack library for **background tasks**
- Works even if:
 - App is closed ○
 - Device restarts
- Used for:
 - Syncing data ○
 - Upload/download
 - tasks ○ Periodic
 - work

Procedure

1. Create a new Android project (Empty Views Activity, Kotlin).
2. Add WorkManager dependency in Gradle.
3. Create a Worker class for background task execution.
4. Create a Repository class to manage WorkManager.
5. Design UI with a Button.
6. Call Repository from MainActivity.

7. Run the app and check Logcat output.

Dependency (build.gradle) implementation

```
"androidx.work:work-runtime-ktx:2.9.0"
```

Worker Class DataSyncWorker.kt

```
import android.content.Context import
androidx.work.Worker
import androidx.work.WorkerParameters

class
DataSyncWorker( context:
Context,          params:
WorkerParameters
) : Worker(context, params)

{ override fun doWork(): Result {

    // Background task simulation
    println("Data syncing in background using WorkManager")

    return Result.success()
}}
```

Repository Class DataRepository.kt

```
import android.content.Context import
androidx.work.OneTimeWorkRequestBuilder import
androidx.work.WorkManager

class DataRepository(private val context: Context)

{ fun syncData() {
    val workRequest = OneTimeWorkRequestBuilder<DataSyncWorker>()
    .build()

    WorkManager.getInstance(context).enqueue(workRequest)
}}
```

UI Layout activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button    android:id="@+id/btnSync"
    android:layout_width="wrap_content"
```

```

android:layout_height="wrap_content"
android:text="Sync Data"/>
</LinearLayout> MainActivity.kt

import android.os.Bundle import
android.widget.Button
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) { super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val repository = DataRepository(this)
        val button = findViewById<Button>(R.id.btnSync)

        button.setOnClickListener
            { repository.syncData()
            }
        }
    }
}

```

Output

- User clicks **Sync Data** button
- WorkManager triggers background task
- Message appears in **Logcat**:

Data syncing in background using WorkManager

